

Web reconnaissance investigation

Authored by Argyro Ouzounoglou. Public portfolio edition.

Internal hunt ID, partial IPs and captured request values are removed.

A password-like request parameter is redacted as a precaution; it is not a verified leaked credential. Timeline correlation alone does not establish causation, attribution or successful compromise.

8 MARCH 2026

THREAT HUNT REPORT

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DISCOVERY OF PRE-ATTACK RECONNAISSANCE PATTERN

PREPARE: DEFINE OF THE HUNT

Hunt Information	Details
Hypothesis	External scanning infrastructure performs reconnaissance against the web application using the /\$(pwd)/ probing pattern in order to identify exposed configuration and development files before launching automated web exploitation attempts.
Threat Hunter Name	Yro Ouzounoglou
Date	8 March 2026
Requestor	Proactive Threat Hunting Initiative
Timeframe for hunt	Mar 4, 2026 - Mar 8, 2026
Threat Hunt Type	Intel-Driven + Hypothesis-Driven Threat Hunt

SCOPING WITH THE ABLE METHODOLOGY

Field	Input
Actor	External automated scanning infrastructure
Behavior	Automated reconnaissance scanning targeting configuration and development files using the /\$(pwd)/ probing pattern followed by automated web exploitation attempts.
Location	Public-facing xxxx web infrastructure (Web traffic logs via Cloudflare Logpush)
Evidence	Cloudflare HTTP request logs showing scanning activity containing the /\$(pwd)/ pattern, followed by subsequent exploitation attempts originating from the same IP infrastructure.

1. CONTEXT

During proactive threat hunting activities on web traffic logs, reconnaissance activity was identified targeting the XXXX **web infrastructure**. The investigation was initiated after reviewing threat intelligence from the [SANS Internet Storm Center](#), which reported scanning activity using a specific probing pattern:

/\${pwd}/

This pattern is commonly associated with **automated scanning activity** attempting to identify exposed configuration files and development artifacts on web servers.

These files may contain sensitive information such as:

environment variables
infrastructure configuration
API keys
application secrets

Examples of commonly targeted files include:

.env
docker-compose.yml
terraform.tfstate
netlify.toml
serverless.yml

If accessible through the web application, **these files may expose sensitive information such as credentials, environment variables, or internal infrastructure configuration.**

2. HYPOTHESIS

Based on the threat intelligence and telemetry observed:

Attackers perform a reconnaissance phase using the /\$(pwd)/ probing technique before launching automated web exploitation attempts.

If this pattern appears in logs, it may indicate:

- pre-attack scanning
- botnet reconnaissance
- automated vulnerability discovery

3. DATA SOURCES

The analysis was performed using Elastic SIEM telemetry from the following dataset:

- cloudflare_logpush.http_request

The investigation focused specifically on the field:

- cloudflare_logpush.http_request.client.request.path

This field contains the **HTTP request path sent by external clients** to the web infrastructure.

4. DETECTION LOGIC

Step 1 IOC identification

Based on intelligence reports, the **following IPs were identified as scanners:**

-
- (observed in our logs at Feb 16, 2026 @ 15:49:42)

These IPs were searched in Elastic.

Step 2 Pattern detection

The following query was used to identify scanning behavior:

```
".*\\V\\$\\(pwd\\).*"
```

This detects requests containing the pattern:

```
/$ (pwd) /
```

Step 3 Scan confirmation

Logs show that the IP performed multiple probing requests such as:

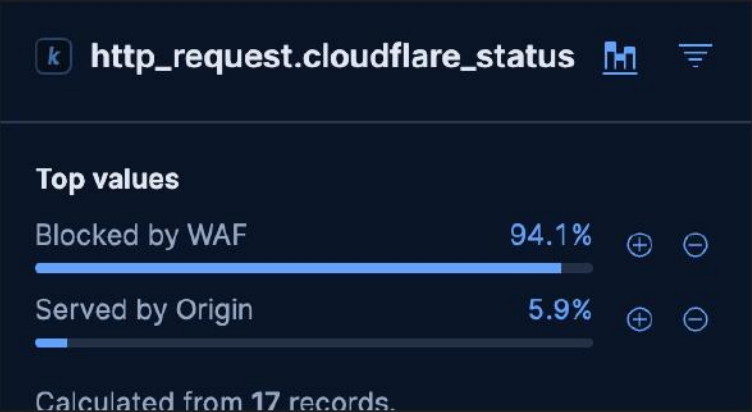
These requests attempt to retrieve configuration files from the application root.

@timestamp per minute	Source IP	URL Original
18:26		/\$ (pwd) /.env
18:26		/\$ (pwd) /.env.development
18:26		/\$ (pwd) /.env.local
18:26		/\$ (pwd) /.env.production
18:26		/\$ (pwd) /.env.staging
18:26		/\$ (pwd) /.git/config
18:26		/\$ (pwd) /.terraform/terraform.tfstate
18:26		/\$ (pwd) /docker-compose.yml
18:26		/\$ (pwd) /netlify.toml
18:26		/\$ (pwd) /next.config.js
18:26		/\$ (pwd) /next.config.mjs
18:26		/\$ (pwd) /package.json
18:26		/\$ (pwd) /serverless.yml
18:26		/\$ (pwd) /serverless.yml
18:26		/\$ (pwd) /terraform.tfstate
18:26		/\$ (pwd) /terraform.tfvars
18:26		/\$ (pwd) /vercel.json

Step 4 WAF interaction

Most of these requests were blocked by the Web Application Firewall:

Status observed:



1 event Served By Origin Status Code 404

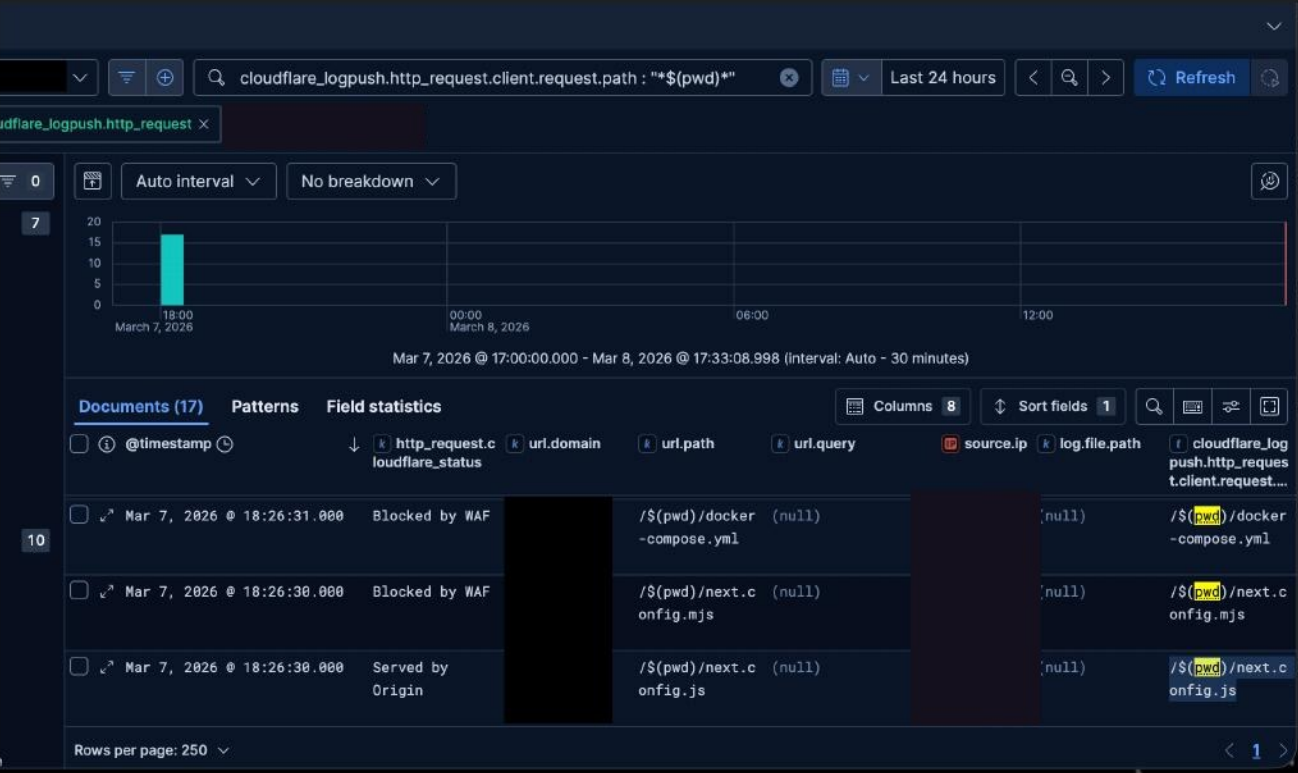
(/\$(pwd)/next.config.js)

This indicates that security controls **prevented most of the probing attempts from reaching the backend server**.

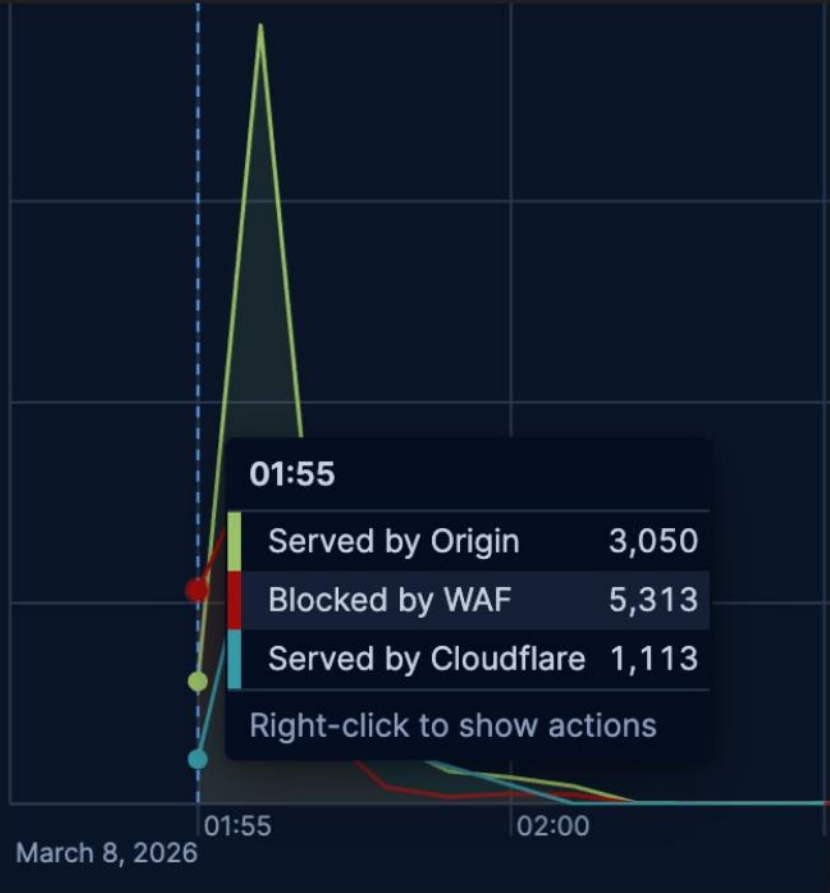
Step 5 Timeline correlation

Investigation revealed the following sequence:

Reconnaissance activity: (Mar 7, 2026 @ 18:26:30 - 18:26:36)



Attack activity detected later: (Mar 8, 2026 @ 01:55:48 - 02:39:26)



This indicates that the attacker performed reconnaissance **several hours before the attack attempt.**

Step 6 Additional malicious traffic

Other IPs were also observed performing requests targeting vulnerable endpoints such as:

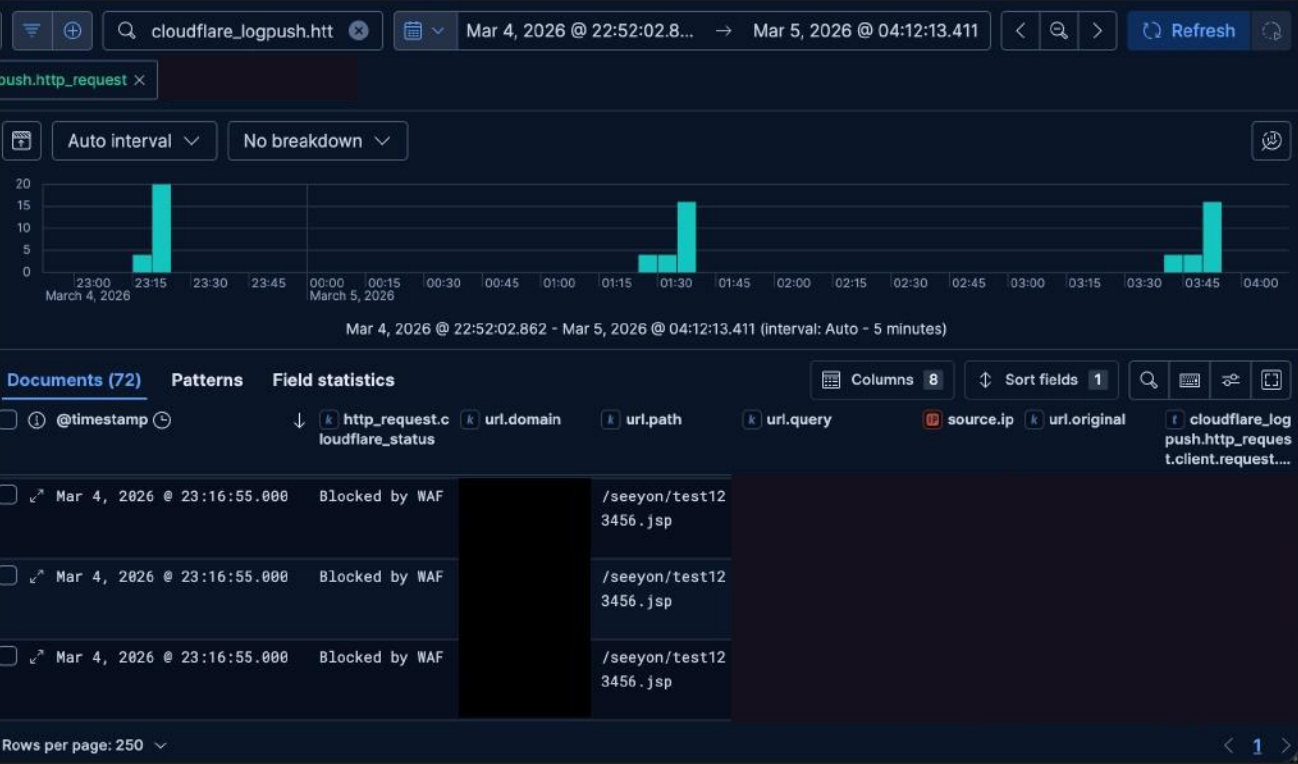
```
/cgi-bin/login_mgr.cgi  
  
/cgi-bin/wapopen  
  
/cgi-bin/GetFileContent.cgi  
  
/seeiyon/test123456.jsp  
  
/pentaho/api/ldap/config/ldapTreeNodeChildren/
```

These patterns correspond to **known web exploitation attempts**, including:

- Command injection attempts
- Authentication bypass attempts
- Remote file access attempts

Alerts were generated shortly after these requests.

Reconnaissance activity: (Mar 4, 2026 @ 23:16:55)



Alert generated [SOC-XXXXX]

Mar 4, 2026 @ 23:20:28.321

Web Server Discovery or Fuzzing Activity

Status

Closed

Risk score

21

Assignees

+

Notes

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Overview

Table

JSON

time

Field	Value
@timestamp	Mar 4, 2026 @ 23:20:28.321

5. DETECTION OPPORTUNITIES

This discovery allows early detection of attacks.

Potential detection logic:

pwd scanning pattern

Elastic KQL:

```
cloudflare_logpush.http_request.client.request.path : "**$(pwd)*"
```

6. MITRE ATT&CK MAPPING

Relevant techniques:

Reconnaissance

- Active Scanning (T1595)

Resource Discovery

- Gather Victim Network Information (T1590)

Initial Access (attempted)

- Exploit Public-Facing Application (T1190)

7. CONCLUSION

The investigation identified a clear attacker workflow:

1. Observed activity indicates a **reconnaissance phase followed by automated exploitation attempts.**
2. The **goal** is to identify exposed configuration or development files.
3. **After reconnaissance,** the attacker performs exploitation attempts against vulnerable web endpoints.
4. Most malicious requests were successfully **blocked by the WAF.**

This discovery highlights an important detection opportunity:

The `/$(pwd)/` scanning activity can serve as an **early indicator of an upcoming web attack.**

Monitoring this pattern **allows defenders to identify malicious actors before exploitation attempts begin.**